Trainings

Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Remove the gear cover and related components. Refer to Procedure 001-031 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-031-tr.html).
- Place the number one cylinder on compression stroke.
- Rotate the engine, using the crankshaft until the index marks on the camshaft gear and the crankshaft gear are pointing at each other.
- Remove the water pump idler gear. Refer to Procedure 001-040 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-040-tr.html).
- Remove the camshaft idler gear. Refer to Procedure 001-036 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-036-tr.html).
- Remove the hydraulic pump idler gear, if equipped. Refer to Procedure 001-039 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-039-tr.html).
- Remove the camshaft gear. Refer to Procedure 001-012 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-012.html).

Remove

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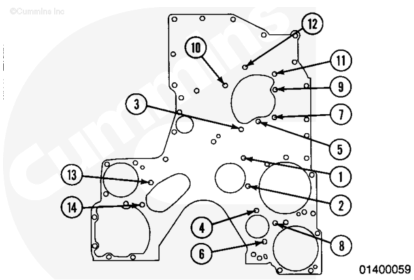


Remove the 14 mounting capscrews from the spacer plate.

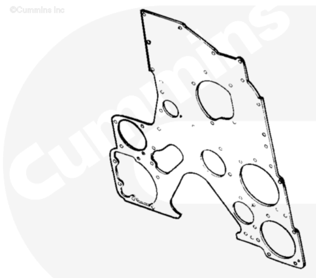
Tag the special capscrews for future identification.

Remove the spacer plate and discard the gasket.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



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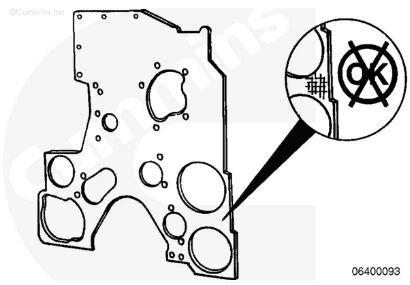
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the gear cover spacer plate. Use solvent.

Dry with compressed air.

Inspect the plate for reuse.



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Check the spacer plate for fretting or damage on both sides of the gasket surface.

Use a feeler gauge and short straight edge to check for plate warpage between adjacent bolt holes on both sides of the spacer plate.

Maximum Warpage Between Adjacent Bolt Holes		
mm		in
0.127	MAX	0.005

Replace the plate if damaged or out of specification.

Check the mounting stud threads for damage.

Install the new studs to the specified height.

Use Loctite™ 577, Part Number 4919202, or equivalent.

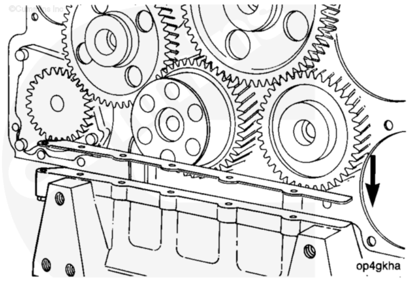
Replace the studs if damage is found.

Accessory Drive Stud (1) - Installed Height			
	mm		in

	31.88	MIN	1.255
	33.15	MAX	1.305

Water Pump Drive Stud (2) - Installed Height			
	mm		in
	62.99	MIN	2.480
	63.75	MAX	2.510

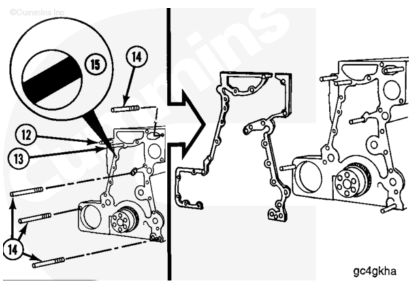
Install



Note : If the engine has the paper style oil pan adapter gasket, remove the pan adapter gasket at perforations. The paper style gasket is designed to be removed at perforation. If equipped, the steel core and metal edge mold gasket can not be cut without causing damage to the pan adapter surface. Inspect the steel core or metal edge mold gasket for reuse.

If applicable, install the paper oil pan adapter service gasket on top of the oil pan adapter. Use spray adhesive to hold it in position.

Apply Cummins® RTV, Part Number 4918882 or 3164067, to the joints of the gasket.



⚠ CAUTION ⚠

Do not use gasket cement on the gasket. Damage to the gasket will result.

Note : Make sure the entire block and gear cover spacer plate surfaces are clean and dry of oil and debris.

If necessary, rotate the diamond dowel (15) so the flat surface is turned facing the master dowel hole at the lower right hand corner of the cylinder block.

This position typically provides the best spacer plate alignment, however, this may **not** be accurate on all engines. In this case, adjust the diamond dowel for the best alignment at the two front corners of the block. The plate should be even with the bottom surface of the block.

Guide studs (14) will aid the installation.

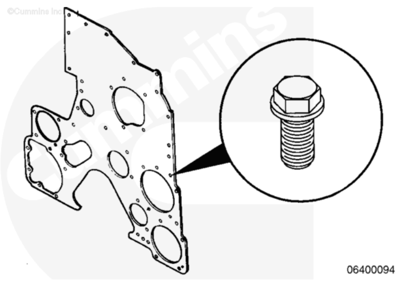
Install the gasket.

Note : If using a paper or steel core gasket, the bottom of the gasket must be even with the bottom of the block and spacer plate.

Note : If using a metal edge mold gasket, the bottom of the gasket will not be even with the bottom of the block. Fill this gap with RTV.

Special capscrews are required to attach the spacer plate and the gear housing.

The capscrews have a captive cone-shaped washer to maintain torque.



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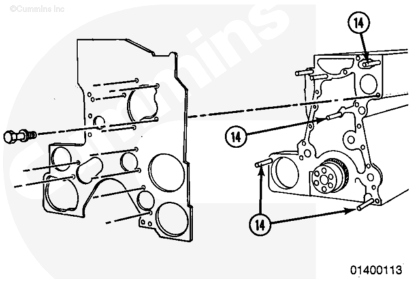
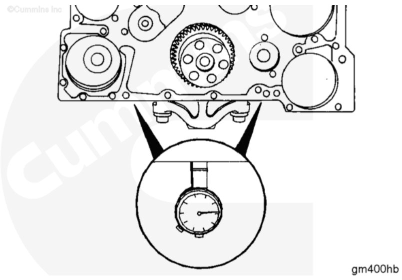


Install the spacer plate and 11 capscrews.

Remove the guide studs (14).

Install the remaining capscrews.

Note : If the oil pan adapter is removed, the spacer plate alignment must be checked. If the oil pan adapter is installed, the spacer plate alignment does not need to be checked.

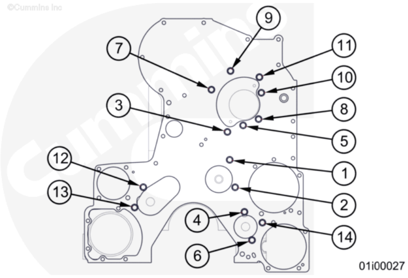


Measure the distance from the bottom of the cylinder block to the bottom of the spacer plate with a depth gauge, Cummins® Part Number ST-547, or equivalent. Measure at the two black arrows as shown.

The bottom of the spacer plate and housing **must** be within 0.05 mm [0.002 in] of the bottom of the cylinder block.

If necessary, align the spacer plate and housing with the bottom of the cylinder block, by turning the diamond dowel. This dowel adjusts the spacer plate alignment. The flat on the diamond dowel typically will face the master dowel for optimal spacer plate alignment. This may **not** be accurate on all engines. In this case, adjust the diamond dowel for the best alignment.

Tighten the capscrews in the sequence shown in the graphic.



Torque Value: 48 n•m [35 ft-lb]

Measure the spacer plate or the housing to the cylinder block alignment again to make sure it is within specifications.

Tighten the capscrews in sequence again.

Torque Value: 48 n•m [35 ft-lb]

Finishing Steps

⚠ WARNING ⚠

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Note : Make sure to check the end clearance on all gears after they have been installed.

- Install the camshaft gear. Refer to Procedure 001-012 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-012.html).
- Install the hydraulic pump idler gear. Refer to Procedure 001-039 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-039-tr.html).
- Install the water pump idler gear. Refer to Procedure 001-040 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-040-tr.html).
- Install the camshaft idler gear. Refer to Procedure 001-036 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-036-tr.html).
- Perform the static injection timing check. Refer to Procedure 006-025 in Section 6 (/qs3/pubsys2/xml/en/procedures/18/18-006-025.html).
- Install the gear cover and related components. Refer to Procedure 001-031 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-031-tr.html).

Last Modified: 06-Feb-2020